1 **(a)** **(i)** State the difference between a scalar quantity and a vector quantity.

.....

..... [1]

(ii) Define momentum.

.....

..... [1]

(b) A test car crashes into a barrier to test the safety features. The test car has a total mass of 950 kg. It is moving with constant velocity from time $t = 0$ for 4.0 s. At $t = 4.0$ s, the car hits the barrier.

Fig. 1.1 shows the car as it hits the barrier.

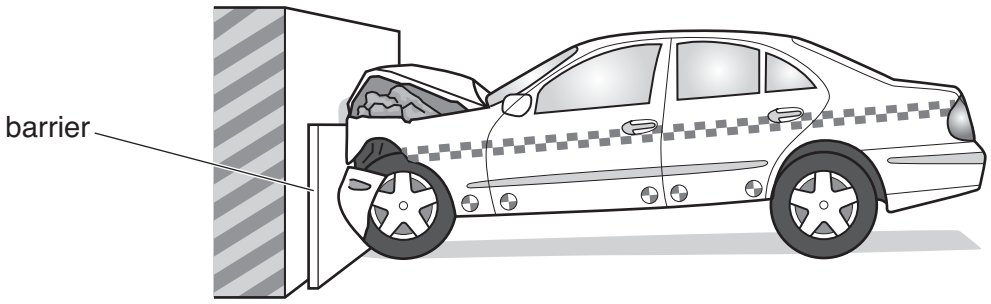

Fig. 1.1

(i) During the test crash, the resultant force acting on the car is 27 000 N. The car takes 1.5 s to come to rest. The deceleration is uniform.

Calculate the initial velocity of the car.

initial velocity = ... [3]

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(ii) On Fig. 1.2, sketch a speed–time graph to show the motion of the car from time $t = 0$ until the car becomes stationary.

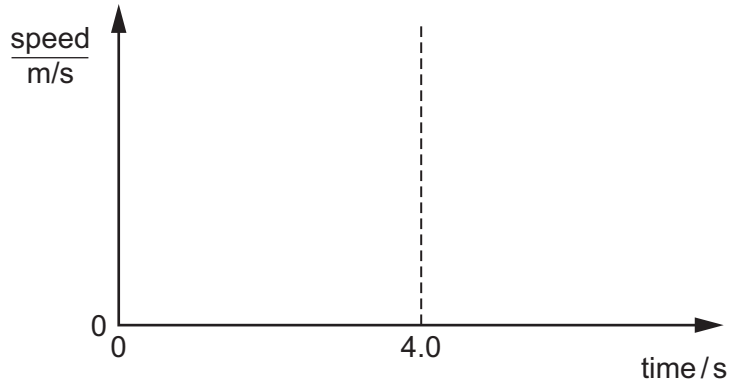

Fig. 1.2

[2]

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